

The Performance of Several Classifiers on the Detection of Fake Negative Hotel Reviews

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Abstract

This paper seeks to compare the performance of several classifiers, namely naive Bayes, logistic regression, classification trees and random forests all both with monograms and bigrams, on recognizing fake negative reviews on the dataset outlined by Ott. et al[7, 6]. The logistic regression classifier has the best performance on this dataset with accuracies of 0.7875 and 0.8125 when using monograms and bigrams, respectively. On the other hand, the naive Bayes is found to have the lowest accuracies, 0.675 and 0.66875 when using monograms and bigrams, respectively. Using only monograms is shown to improve the results when using classification trees and random forest classifiers, while using bigrams as well improves the average results of naive Bayes and logistic regression approaches slightly. Finally an analysis will be provided on the most important words used to distinguish fake reviews from real ones in the used dataset.

1 Introduction

Online reviews have become an important tool for consumers to decide which products they want. As such, dedicated platforms such as Tripadvisor[9] and many others have begun providing centralized platforms for consumers to compare products based primarily on customer reviews.

Unfortunately, this has resulted in business owners having an interest in the reviews for their products being positive, and their competition's being negative. As such, fake reviews are sometimes posted to such platforms in order to achieve this. For the sake of integrity and honesty to the customer, it is naturally desirable to remove such fake reviews. Discerning between real and fake reviews is a difficult manual process according to Ott et al.[7]. The research shows that human reviewers achieved a maximum accuracy of approximately 62%. In addition, the vast amount of review data that is generated makes it impossible to do manual verification. For example, Tripadvisor has received 26 million reviews and has taken down nearly a million fake reviews in 2020 alone[2].

This paper investigates automated methods to discern between real and fake reviews. For this purpose, we will investigate the potency of four automatic classification approaches: Naive Bayes, Logistic Regression, Classification Trees and Random Forest. We will look into which method among these performs best when trying to classify reviews from a dataset of known fake and genuine reviews, and if this is good enough for practical applications. These classification methods will be analyzed on both the monograms and bigrams found in the dataset. We will only consider the negative reviews, in order to limit the scope of this paper.

To do this, the dataset will be described in more detail in section 2. The approach used and results found will be described in section 3, after which the findings will be analysed in more detail in section 4.

2 Experiment Data

The used dataset are the negative reviews from the dataset presented by Ott. et al[7, 6]. This data consist of 800 reviews, of which 400 are truthful and 400 are deceptive. All reviews have a minimum length of 150 characters and are about hotels in Chicago.

2.1 How the data is obtained

The reviews are sampled in such a way that there are exactly 20 reviews for each of Chicago's most popular hotels. The data for the deceptive reviews is generated by hiring workers in order to write a review for a competitors hotel, where each review is written by a unique worker that lives in the United States. For this purpose each worker is provided with the link and the hotel. In order to ensure the quality of these reviews, a human judge verifies that it is bigger than 150 words, conveys an overall negative experience and is of sufficient quality.

The truthful reviews are sampled from Expedia, Hotels.com, Orbitz, Priceline, TripAdvisor, and Yelp. Even

though these reviews might not be 100% truthful, research suggest that the percentage of deceptive reviews is relatively small[3, 5]. The truthful reviews are then truncated at 150 words. After which they are sampled with a log-normal distribution fit to the lengths of the deceptive reviews in order to achieve a similar distribution as the negative reviews.

2.2 The provided experiments

There are some experiments done on this data, some of which will be used in this paper. The paper outlining this dataset contains the recall, precision, accuracy and F1 score of three human judges. The paper accompanying this dataset also gives the performance of several automatic classifiers, but because the feature selection is different only the data concerning the human judges will be used as a benchmark for comparisons in subsection 3.3.

3 Experimental Setup and Results

This section will present the different methods employed to identify reviews as genuine or false, as well as the results of these methods and their analysis.

3.1 Experimental Setup

There are a number of classification methods that will be compared in this paper. These methods are multinomial naive Bayes, logistic regression (with a Lasso penalty), classification trees and random forests. These methods can be used with either the monograms or the set of both bigrams and monograms. As such, there are eight total classification methods to be tested.

In order to get the best results when applying a method to the data, we first apply preprocessing to it. This preprocessing reduces the tests to a number of parameters more easily usable by the methods. The preprocessing consists of a number of steps:

1. All characters are converted to lowercase.
2. All words are tokenized; all punctuation except ellipses ('...') is removed.
3. The NLTK lemmatizer[4] based on Wordnet is used to lemmatize all words.
4. The NLTK corpus for (English)[4] is used to remove stopwords.
5. A corpus of all monograms and bigrams now present in the texts is generated. The feature vector of each text becomes, for each word from the corpus, whether or not that word occurs in that text.
6. The Mutual Information (MI) of all features is calculated. Only the features with the best MI score are kept while the rest are discarded. How many features are kept is a parameter that is to be optimized as well as the hyperparameters of the classifiers.
7. The sklearn implementation of tf-idf[1] may be applied to update the feature vectors to contain the weighted frequencies rather than the raw frequencies of words. This means that the impact of common words that are not very informative will be lower in the feature vectors, whereas words that are more informative will be weighted more highly. Whether or not tf-idf is used is a hyperparameter to be tuned.

Then, finally, the classifiers can be applied. For this purpose, we want to find the combinations of hyperparameters and number of selected features that yield the best results. To do this, we apply an iterative search method; we first test each classifier with a number of combinations of hyperparameters and see which gives the best results. Then, we test more combinations spread more finely around the best combination of hyperparameters found in the last iteration. This is repeated until no further improvements can be found by iterating further.

The gridsearch method of sklearn is used to perform the hyperparameter tuning. This method uses cross validation over 5 folds to compute the score of the hyperparameters. Using this approach reduces overfitting on the training set.

Finally, once the results of each classification method with the best combination of hyperparameters is known, they can be analysed. The results of the classifications methods will be presented in subsection 3.2, and analysed in subsection 3.3

3.1.1 Hyperparameters & Tuning

A number of hyperparameters have been tuned to find the best results on the training set. A number of parameters occur for each of the methods. These are 'document frequency', 'number features' and 'use tf-idf'. These denote the minimum number of reviews in which a word must occur before it can be used as a feature, how many features the model is trained on and whether or not tf-idf is used, respectively. For the number of features used, the used features are simply the specified number of features that have the highest mutual information score.

Furthermore, a number of features are also specific to each classifier. Naive Bayes uses the feature 'alpha' which specifies the smoothing parameter of the model. Logistic Regression uses the parameters 'C' and 'solver', denoting the inverse regularization strength and the solver used to find solutions, respectively. Classification trees use the parameter 'ccp alpha', the cost-complexity pruning parameter. Random forest models furthermore also use 'n estimators' and 'max features', the number of trees in the forest and the maximum number of features considered at each node, respectively.

The hyperparameters that were found to optimize the accuracy on the test set during hyperparameter tuning are listed in tables 9 to 12.

3.2 Experimental Results

For each of the classification methods, the hyperparameters have been tuned to find the best results, both with and without bigrams. An overview of the results of using these models on the test dataset are presented in this subsection. All classification methods have been repeated eleven times in order to reduce the effect of random feature selection. The confusion matrix of the results with the median accuracy are shown in tables 1 to 4.

Random Forest confusion matrices

		Predicted class		Total			Predicted class		Total
		Deceptive	Truthful				Deceptive	Truthful	
True class	Deceptive	53	27	80	True class	Deceptive	38	42	80
	Truthful	4	76	80		Truthful	5	75	80
Total		57	103	160	Total		43	117	160

Table 1: The left table shows the confusion matrix of Random Forest using monograms. The table on the right shows the confusion matrix of Random Forest using bigrams

Naive Bayes confusion matrices

		Predicted class		Total			Predicted class		Total
		Deceptive	Truthful				Deceptive	Truthful	
True class	Deceptive	45	35	80	True class	Deceptive	40	40	80
	Truthful	17	63	80		Truthful	13	67	80
Total		62	98	160	Total		53	107	160

Table 2: The left table shows the confusion matrix of Naive Bayes using monograms. The table on the right shows the confusion matrix of Naive Bayes using bigrams

Classification Tree confusion matrices

		Predicted class		Total			Predicted class		Total
		Deceptive	Truthful				Deceptive	Truthful	
True class	Deceptive	59	21	80	True class	Deceptive	43	37	80
	Truthful	20	60	80		Truthful	14	66	80
Total		79	81	160	Total		57	103	160

Table 3: The left table shows the confusion matrix of Classification Tree using monograms. The table on the right shows the confusion matrix of Classification Tree using bigrams

Logistic Regression confusion matrices

		Predicted class		Total			Predicted class		Total
		Deceptive	Truthful				Deceptive	Truthful	
True class	Deceptive	58	22	80	True class	Deceptive	58	22	80
	Truthful	12	68	80		Truthful	8	72	80
	Total	70	90	160		Total	66	94	160

Table 4: The left table shows the confusion matrix of Logistic Regression using monograms. The table on the right shows the confusion matrix of Logistic Regression using bigrams

The accuracy, precision, recall, and F1 score are calculated using the confusion matrices. The formulas used are the following. Here, 'TP' denotes True Positive and 'TN' denotes True Negative. Deceptive is the positive class in the paper.

$$Accuracy = \frac{TP + TN}{TP + FP + TN + FN} \quad (1)$$

Accuracy describes the fraction of the predictions that is classified correctly.

$$Precision = \frac{TP}{TP + FP} \quad (2)$$

Precision describes the fraction of deceptive reviews in the set of reviews with a correct prediction label.

$$Recall = \frac{TP}{TP + FN} \quad (3)$$

Recall describes the fraction of truthful reviews in the set of reviews with a correct prediction label.

$$F1 \text{ score} = \frac{2 * Precision * Recall}{Precision + Recall} \quad (4)$$

F1 Score is a scoring metric which takes both precision and recall into account. Both of the values need to be relatively high to result in a high F1 score. This prevents a classifier from, for example, only assigning cases to the positive value to optimize the precision. The range of the F1 score is zero (in case a classifier does poorly) to one (in case it does very well).

3.3 Statistical Analysis of Results

A one-tailed binomial distribution is used in order to test whether or not (results of) classification methods are significantly better or worse than others. The null hypothesis, that one classification method is better or worse than another, will be rejected if the probability to get that result or better/worse is great or equal than 95%. In the next section the results will be compared with three human judges trying to discern fake and real reviews, as outlined in Ott et al.[7]. Furthermore, the results from different methods will be compared to each other in order to see which methods differ significantly from one another. We will also look at which features are the most indicative of a review being genuine or deceptive.

3.3.1 Comparison against human judges

The quality of classification by human judges as described in [7] can be seen in Table 5. We can furthermore compare the quality of classification by each judge to each method by means of the aforementioned single-tailed binomial distribution. When we apply this method, we find the probability that each classification method is more accurate than each judge. The results of this comparison are shown in Table 6.

	Accuracy (%)	Precision (%)	Recall (%)	F1 score (%)
JUDGE 1	61.9	57.9	87.5	69.7
JUDGE 2	56.9	53.9	95.0	68.8
JUDGE 3	53.1	52.3	70.0	59.9

Table 5: Accuracy, Precision, Recall and F1 score of classification by human judges.

Judge	NB		LR		CT		RF	
	monogram	bigram	monogram	bigram	monogram	bigram	monogram	bigram
Judge 1	0.923	0.89524	1	1	0.99966	0.94491	1	0.98922
Judge 2	0.99685	0.99475	1	1	1	0.99816	1	0.99985
Judge 3	0.99989	0.99979	1	1	1	0.99995	1	1

Table 6: The probabilities that each method discern between real and fake reviews more accurately than each human judge. Here 'RF' is used to denote 'Random Forests', 'NB' denotes 'Naive Bayes', 'CT' denotes 'Classification Tree' and 'LR' denotes 'Logistic Regression'. The second term in each header indicates if this method uses only monograms or bigrams as well.

The results in Table 6 show that all classifiers score a higher accuracy on the test set than the judges. Almost all of the classifiers are better than the judges with a probability bigger than 95%. The only exceptions are either naive Bayes classifier and the classification tree using bigrams as these have a chance less than 95% to be better than judge 1. It can therefore not be concluded that all classifiers are better than all judges. However, it can be said that all classifiers are better than judge 2 and 3 and that the logistic regression, random forest and classification tree (using only monograms) classifiers are better than judge 1 with a 95% probability.

3.3.2 Comparison against other methods

Finally, we look to compare the various methods against each other to find out which is most effective. Using the one-tailed binomial distribution we can find the probability that any one method is better (or worse) than any one other method. The results of these comparisons are shown in Table 13 and Table 14. The accuracy, precision, recall and F1-score of each method is listed in Table 15,

We note that we can conclude that one method outperforms another with a p-value of 0.05 when the probability it is better is at least 0.95 and the probability it is worse is at most 0.05. Thus, when looking at this data, a number of things can be observed. We can see that the probability that either discriminative linear model (Logistic Regression with monograms or bigrams) is better than either generative model (Naive Bayes with monograms or bigrams) is at least 0.99938 and at most 0.99999. Similarly, the probability that either logistic regression model is worse than either naive Bayes model is at most 0.00033. As such, since these probabilities are greater than 0.95 and lower than 0.05, respectively, we can conclude that a generative linear model is an improvement over a discriminative linear model with $p = 0.05$. This is further supported by the data in Table 15, which shows that either logistic regression method outperforms either naive Bayes approach in every metric.

When adding the random forest to the comparison, we notice that the probability that a random forest with monograms is better than naive Bayes is always over 0.99. Similarly, the probabilities that this method is worse are mere fractions of a percent. As such, we can confidently conclude that a random forest with monograms outperforms naive Bayes. For a random forest with bigrams we are unable to draw such conclusions by the aforementioned criteria. The results in Table 15 do suggest the random forest outperforms naive Bayes, but such conclusion cannot definitively be drawn with the required p-value.

Comparing the random forest classifiers with the logistic regression classifiers using the previously described method only allows us to confidently draw the conclusion that the random forest approach with bigrams is worse than either logistic regression approach. Looking at the metrics in Table 15, we can indeed see that random forest with monograms is comparable to both logistic regression models while random forest with bigrams produces worse accuracy, precision and F1-scores, but a better recall. As such, the random forest with bigrams is generally worse than logistic regression methods, but could be better suited when a high recall is preferred.

When comparing the methods with monograms to those with bigrams, we see that including bigrams never improves the results of a method over those using just the monograms to a degree where statistically significant conclusions can be drawn. In fact, only when using naive Bayes or logistic regression do we see an improvement caused by the addition of bigrams in the various metrics. Still, we are not able to draw a conclusion about this with sufficient confidence. On the other hand, we can conclude with enough certainty that adding bigrams to the random forest and classification trees worsens the results. It seems likely that including bigrams as features increases the amount of overfitting done by some methods, as bigrams may occur less frequently in texts than monograms, leading for the methods to optimize for specific terms that are less likely to be found.

As such, we see that logistic regression seems to yield better results than naive Bayes. The results of the former are comparable to those of classification trees. Including bigrams only manages to improve the results of naive Bayes and logistic regressions a little on average, and not enough such that conclusions can be drawn with a p-value of 0.05. We can, however, conclude with that confidence that including bigrams makes the results of the classification trees and random forests worse.

3.3.3 Most discerning features for deceptive reviews

To retrieve the most important words that distinguish a deceptive review, we take the monograms and bigrams with the highest mutual information score in sorted order. The top five words that are more common in deceptive reviews are the most discerning features for deceptive reviews. To get the list of the five most important words the minimum document frequency is 10. As such, each word listed below must have occurred in at least ten reviews.

The most discerning features of deceptive reviews

Monograms			Bigrams		
Word	Deceptive	Truthful	Word	Deceptive	Truthful
Chicago	246	113	Chicago	246	113
Smell	75	13	Smell	75	13
Luxury	38	3	Luxury	38	3
Decide	57	14	Hotel Chicago	54	11
Recently	49	10	Decide	57	14

Table 7: The most discerning features for a deceptive review, and the number of deceptive and truthful reviews which contains these words.

As seen in Table 7, the list of the most important features changes little when including bigrams as well as monograms. It is observed that in deceptive reviews there is a strong bias to mention the hotel location and name. This could be because it is difficult to come up with a large number of fake reviews and make them distinct, because this is some of the (relatively little) information available to the reviews author or because, since the goal is to create a negative association with the hotel, it is important to mention it. In bigrams it shows that not only the place of the hotel is used but also the hotel name (which often includes 'Chicago', as the hotels are all in Chicago) is mentioned more often in deceptive reviews. When looking at the data this bias it is observed that many of the deceptive reviews start with mentioning the hotel name. An example of this is: "I recently stayed at the Hotel Allegro Chicago with my wife. ...". Here, the authors goal could be to give negative associations with the hotel name.

Furthermore, a lot of deceptive negative reviews complain about the smell or luxury of the hotel. Which is hardly mentioned in truthful negative reviews. This indicates that the true issues customers have with the hotel is different.

The reason that the words "decide" and "recently" are so much more frequent in fake reviews is more difficult to uncover. The word "recently" might try to make it look like the negative aspects of the hotel are still present, but it is not clear from the data.

In any case it is clear that the presence of these features strongly increases the probability that a review is disingenuous. The exact reason why people writing fake reviews tend to use these terms more often is left to future research as researching this is outside the scope of this paper.

3.3.4 Most discerning features for truthful reviews

To select the five most important words for discerning truthful reviews, same process is used as in subsection 3.3.3. Here, however, the five features most instrumental in identifying truthful reviews instead of those for deceptive reviews.

The most discerning features of truthful reviews

Monograms and Bigrams		
Word	Truthful	Deceptive
Location	96	31
Elevator	47	10
Great	87	40
Open	48	15
Priceline	14	0

Table 8: The most discerning features for a deceptive review, and the number of deceptive and truthful reviews which contains these words. When using bigrams as well as monograms, the list of the five best features remains the same as when using just monograms.

As seen in Table 8, the guests of these hotels in Chicago are mostly negative about the location of their hotels and the elevators. Only guests that stayed at the hotels could know about this seemingly common problem.

It is interesting to see that a lot of truthful negative reviews mention positive 'great' things about the hotel. This is apparently a good indicator for whether someone is trying to be deceptive in their review.

Priceline[8] is a booking site used in the United States to find cheap hotels. When looking through the reviews most reviews contained negative remarks about Priceline. It seems to be specific to Chicago and thus only people that actually booked the hotel through that site had this negative experience, whereas people who did not actually visit the hotels would not share this experience.

The word "open" is difficult to explain since the word can have a lot of meanings in a negative reviews, for example closed bar with a plate stating open on the door, a hotel room being opened by the cleaner without knocking or other situations. So there is no clear explanation why this would be more common in truthful reviews.

In any case, it is clear which words indicate a review is genuine most strongly. As with the deceptive features, why these words tend to be used less by people writing deceptive reviews is left for future research.

4 Conclusion

It has been shown that the classifiers consistently perform better than the manual screening for fake reviews by the judges. As such, automatized detection proves to be a promising prospective method for detecting deceptive reviews.

Out of the four tested classification methods, it is shown that Logistic Regression with bigrams and Random Forests without bigrams yield the best results. For the linear classification methods, using the bigrams as well as the monograms increases the quality of the results on average. On the other hand, using only the monograms improves the results of the Classification Tree and Random Forest methods. The Naive Bayes approach performs the worst out of the four classifiers, but still manages to achieve better results than any of the human judges.

When identifying the most discerning features of deceptive and truthful reviews, many of these terms are reasonable to expect in such reviews. This validates the found models, and thus solidify confidence in our findings. However, further, more extensive research into the specific terms falls outside of the scope of this paper.

References

- [1] sklearn community. *sklearn.feature_extraction.text.TfidfTransformer*. visited on 04-11-2022. URL: https://scikit-learn.org/stable/modules/generated/sklearn.feature_extraction.text.TfidfTransformer.html.
- [2] R. Hart. *Tripadvisor Took Down Nearly 1 Million Fake Reviews Last Year*. visited on 04-11-2022. Oct. 2021. URL: <https://www.forbes.com/sites/roberthart/2021/10/27/tripadvisor-took-down-nearly-1-million-fake-reviews-last-year/?sh=2c1533102fa0>.
- [3] D. Mayzlin, Y. Dover, and J. Chevalier. "Promotional reviews: An empirical investigation of online review manipulation". In: *American Economic Review* 104.8 (2014), pp. 2421–55.
- [4] NLTK Team. *NLTK :: Natural Language Toolkit*. visited on 04-11-2022. URL: <https://www.nltk.org>.
- [5] M. Ott, C. Cardie, and J. Hancock. "Estimating the prevalence of deception in online review communities". In: *Proceedings of the 21st international conference on World Wide Web*. 2012, pp. 201–210.
- [6] M. Ott, C. Cardie, and J. T. Hancock. "Negative deceptive opinion spam". In: *Proceedings of the 2013 conference of the north american chapter of the association for computational linguistics: human language technologies*. 2013, pp. 497–501.
- [7] M. Ott, Y. Choi, C. Cardie, and J. T. Hancock. "Finding Deceptive Opinion Spam by Any Stretch of the Imagination". In: (2011). DOI: 10.48550/ARXIV.1107.4557. URL: <https://arxiv.org/abs/1107.4557>.
- [8] Priceline.com - *The Best Deals on Hotels, Flights and Rental Cars*. visited on 2022-11-4. URL: <https://www.priceline.com>.
- [9] Tripadvisor. *Tripadvisor*. visited on 04-11-2022. URL: <https://www.tripadvisor.nl/>.

5 Appendix

	Monograms	Bigrams
document frequency	4	4
number features	611	636
alpha	0.032	0.18
use tf-idf	yes	yes
training accuracy	0.934	0.973

Table 9: Parameters for Naive Bayes found during hyperparameter tuning.

	Monograms	Bigrams
document frequency	5	10
number features	158	264
C	5.1	8.2
solver	saga	saga
use tf-idf	no	no
training accuracy	0.872	0.856

Table 10: Parameters for Regularized Logistic Regression found during hyperparameter tuning.

	Monograms	Bigrams
document frequency	3	10
number features	9	11
ccp alpha	0.0001	0.008
use tf-idf	yes	yes
training accuracy	0.783	0.770

Table 11: Parameters for Classification Trees found during hyperparameter tuning.

	Monograms	Bigrams
document frequency	4	8
number features	300	360
n estimators	285	250
max features	1	1
ccp alpha	0.004	0.002
use tf-idf	yes	yes
training accuracy	0.908	0.915

Table 12: Parameters for Random Forest during hyperparameter tuning.

The probabilities that one model is better than another

	RF		NB		CT		LR	
	monogram	bigram	monogram	bigram	monogram	bigram	monogram	bigram
RF monogram	-	0.99846	0.99994	0.99997	0.96817	0.99988	0.69679	0.38931
RF bigrams	0.00148	-	0.78396	0.83042	0.12882	0.73067	0.00802	0.00078
NB monogram	0.00007	0.17696	-	0.53751	0.02423	0.4037	0.00061	0.00003
NB bigram	0.00004	0.137	0.40403	-	0.01625	0.3402	0.00034	0.00002
CT monogram	0.02578	0.84048	0.96913	0.97935	-	0.95498	0.08497	0.01613
CT bigram	0.00014	0.22393	0.53784	0.60409	0.03536	-	0.00106	0.00007
LR monogram	0.25267	0.99038	0.99938	0.99967	0.8938	0.99886	-	0.19346
LR bigram	0.54861	0.99924	0.99998	0.99999	0.98049	0.99995	0.76405	-

Table 13: Here 'RF' is used to denote 'Random Forests', 'NB' denotes 'Naive Bayes', 'CT' denotes 'Classification Tree' and 'LR' denotes 'Logistic Regression'. The second terms indicate if this method uses only monograms or bigrams as well.

The probabilities that one model is worse than another

	RF		NB		CT		LR	
	monogram	bigram	monogram	bigram	monogram	bigram	monogram	bigram
RF monogram	-	0.00084	0.00003	0.00002	0.02062	0.00006	0.23871	0.53173
RF bigram	0.9973	-	0.16958	0.1302	0.82988	0.21604	0.98687	0.99852
NB monogram	0.99985	0.77495	-	0.39655	0.96383	0.5297	0.99886	0.99993
NB bigram	0.99992	0.82185	0.5297	-	0.97513	0.59598	0.99935	0.99996
CT monogram	0.96028	0.12055	0.02065	0.01348	-	0.03087	0.88172	0.97422
CT bigram	0.99971	0.72136	0.3959	0.33274	0.94851	-	0.99806	0.99986
LR monogram	0.68032	0.00587	0.00033	0.00017	0.07612	0.00063	-	0.74732
LR bigram	0.37342	0.0004	0.00001	0.00001	0.01218	0.00002	0.17998	-

Table 14: Here 'RF' is used to denote 'Random Forests', 'NB' denotes 'Naive Bayes', 'CT' denotes 'Classification Tree' and 'LR' denotes 'Logistic Regression'. The second terms indicate if this method uses only monograms or bigrams as well.

Accuracy, precision, recall and F1-score of the models on the test set

Measure	RF		NB		CT		LR	
	monogram	bigram	monogram	bigram	monogram	bigram	monogram	bigram
Accuracy	0.80625	0.70625	0.675	0.66875	0.74375	0.68125	0.7875	0.8125
Precision	0.73786	0.64103	0.64286	0.62617	0.74074	0.64078	0.75556	0.76596
Recall	0.95	0.9375	0.7875	0.8375	0.75	0.825	0.85	0.9
F1-Score	0.8306	0.7614	0.7079	0.7166	0.7453	0.7213	0.8000	0.8276

Table 15: Accuracy, precision, recall and F1-score of the median run of the models on the test set. Here 'RF' is used to denote 'Random Forests', 'NB' denotes 'Naive Bayes', 'CT' denotes 'Classification Tree' and 'LR' denotes 'Logistic Regression'. The second terms indicate if this method uses only monograms or bigrams as well.